ANDREW FISCHER

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Summary

Fourth-year Electrical and Computer Engineering student with hands-on experience in embedded systems and hardware development. Three years of academic and professional experience in C/C++ programming, mixed-signal circuit design, PCB layout, assembly, and hardware debugging. Currently an intern at Rocket Lab.

Skills

Embedded Systems: Arduino (AVR), STM32 (ARM Cortex-M), Raspberry Pi Pico (RP2040, ARM Cortex-M0+)
Interfaces: CAN, UART, USB, SPI, I2C; RTOS: FreeRTOS
Languages & Tools: C, C++, Python, JavaScript, MATLAB, Lua, ROBOT, Verilog, VHDL, Bash, CMake, Git, TeamCity
Design Software: KiCAD, Altium, AutoCAD, LTspice
Lab Equipment: Logic Analyzer, Oscilloscope, Digital Multimeter, Signal Generator
Qualifications: Licensed Amateur Radio Operator - Basic and Advanced (Honours)

Work Experience

Spacecraft Operations Automation Intern — MAY. 2026 – Present
Rocket Lab — Auckland, NZ

- Developed SITL & HITL (FlatSats) tests using Robot Framework to verify end-to-end interactions between test infrastructure and flight software, including telemetry validation, command execution, and FDIR scenarios.
- Developed browser-based JavaScript tools to streamline day-to-day engineering operations, including CRC32/MD5 verification and data conversion utilities.
- Contributed to CI/CD pipelines for automated build, test, and validation of flight software using continuous integration workflows within ITAR/CUI-controlled environments, collaborating with flight and ground software teams.
- Developed Python scripts to aggregate telemetry packets and automate data-budget calculations across various TT&C modes.

Electrical Engineer (Internship) JAN. 2025 – AUG. 2025 (8 MO)
ContainerWest Richmond, BC

- Independently designed, assembled and documented a working TRL-6 expandable annunciator panel, enabling firefighters to interact with the system to simulate events; implemented mixed-signal PCB design in KiCAD and embedded firmware in C++.
- Developed ladder logic programs for OMRON NX1P2 PLCs to automate tasks and monitor live fire burn props to ensure safe operation.
- Gained practical hands-on experience with high-power equipment, including three-phase AC induction motors, incinomites, and transformers.
- Collaborated with tradespeople and electricians, providing technical guidance and support for installations and troubleshooting.
- Developed ENEFEN-approved system flow graphs and electrical schematics in AutoCAD and performed the associated on-site installations and control system modifications.
- Performed independent site visits and troubleshooting, working directly with clients to ensure reliable and safe operation of the live fire burn props.

Technical Projects

Supercapacitor Bike

- Designed a complete embedded system to drive a modified BLDC motor.
- Developed lightweight, high-performance ATmega328P firmware, including I2C, SPI, EEPROM, Timer and PWM drivers using C and AVR-ASM.
- See project here.

Education

University of Victoria Bachelor of Engineering, Electrical & Computer Engineering, 4th year (CGPA : 3.7/4.0)